
Exchange Fees and Overall Trading Costs[†]

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Abstract

In the US market, two popular exchange fee structures are “maker-taker” and “inverted” (also known as “taker-maker”). Maker-taker exchanges charge fees for taking liquidity and give rebates for providing liquidity, while inverted exchanges give rebates for taking liquidity and charge fees for providing liquidity. Using live trade data from an institutional investor that applies a flexible trading approach and seeks to participate in the natural market liquidity, we find that net of fees and rebates, trading costs are statistically indistinguishable between maker-taker and inverted exchanges. This finding has important implications for venue routing decisions in trading algorithms that do not demand immediacy, suggesting that the optimal routing strategy in terms of the tradeoff between cost and volume per unit of time is a pro-rata allocation across venues. It is also relevant to the recent discussions on legislation to fix fees across all trading venues.

1. Introduction

Historically, the NYSE, NASDAQ, and AMEX stock exchanges matched buy and sell orders and charged access fees on both sides when an execution occurred. During the late 1990s, the number of trading venues in the US market proliferated. This change resulted in a considerable increase in options of where to trade: a broker can now trade a company’s stock on 13 licensed exchanges and more than 50 off-market venues, or “dark pools.” When brokers are routing an order, they must consider each of these different venues to ensure the best trade execution for their clients. To attract trading interest, venues employ a range of tools, such as rebates and fees, to incentivize traders to execute on their respective platforms. “Maker-taker” exchanges charge fees to liquidity takers and give rebates to liquidity providers, while “inverted” exchanges give rebates to liquidity takers and charge fees to liquidity providers. One way exchanges profit is from the difference between the fees they charge and the rebates they provide.¹

The Order Protection Rule ensures investors receive the best quoted price across venues when executing trades.² However, even when quoted prices are the same, net prices across venues can be different if fees are taken into consideration. That is, if quoted prices are held constant, a liquidity taker would receive the cheapest net price by transacting at a venue that offers rebates to liquidity takers while liquidity providers are incentivized to route orders towards venues offering rebates to liquidity providers.³

As a result, the fee structure across exchanges creates a preferential ordering across venues based on the net price of a security and is likely to have wide implications for market functioning. Because

¹ In 2005, the SEC’s Regulation National Market System (Reg NMS) adopted Rule 610(c) that restricted fees to be no more than \$0.0030 per share.

² In 2005, Reg NMS adopted Rule 611 (also known as the “Order Protection Rule” or the “trade-through rule”), which provided intermarket price priority for displayed and accessible quotations.

³ See Battalio, Corwin, and Jennings (2016).

of this ordering, for instance, maker-taker venues are likely to attract liquidity providers and exhibit higher market impact, while the liquidity seekers might migrate towards inverted venues.

Specifically, while venues with the largest liquidity rebates are likely to attract the most liquidity, these rebates are going to be offset by higher taker fees. Brokers placing market orders might be incentivized to route orders away from exchanges with the most liquidity but high taker fees, and instead route to venues that offer rebates or charge small (or zero) access fees for takers. This outcome might not be the optimal choice for clients as the brokers might miss opportunities for execution against hidden liquidity on maker-taker venues. Routing away from maker-taker venues could also negatively impact investors if brokers are able to execute only a small quantity in an inverted venue, leaving a residual that has to be routed to another venue and, as a result, increases the potential for information leakage and adverse price moves before the completion of the order (Barbon *et al.*, 2019).

Similarly, brokers searching for the highest liquidity rebate might put less weight on other considerations, such as a potentially lower probability of execution and a higher probability of adverse selection. This is likely to happen because, as shown by Battalio, Corwin, and Jennings (2014) and Wah, Feldman, Chung, Bishop, and Aisen (2017), a negative relation exists between several measures of limit order execution quality and rebate/fee level, i.e., non-marketable⁴ limit orders routed to venues with lower/negative liquidity rebates are executed with faster rates and less adverse selection than orders routed to venues with higher liquidity rebates.

Regulators have also recently inquired about a possible conflict of interest between brokers and clients in a maker-taker system and a lack of transparency due to the quoted bids and offers not representing actual net trading prices. To empirically assess the importance of these issues, the US Securities and Exchange Commission (SEC) in 2018 announced an experiment that would divide US-listed stocks into three groups, with each test group employing different levels of access fees and rebates.⁵ Such an experiment quickly attracted the attention of the industry, with fervid support from some and staunch opposition from others. For instance, the IEX Group Inc., whose exchange doesn't offer rebates, said the SEC's move is a "monumental win for investors and other stakeholders who have pushed for a fairer and less conflicted market throughout the years."⁶ However, a number of exchanges have sued the SEC to halt the pilot, which in their words is "intrusive, ill-conceived and likely to harm the equities markets." The NYSE's filing called the pilot "arbitrary and capricious" and said it "exceeds the Commission's authority."⁷

The lack of clarity on the total net effects of the current fee structure has spurred a heated debate among practitioners and academics. Theoretically, the effects of fees on quotes, market impact, and costs are indeed unclear. For instance, Angel, Harris, and Spatt (2011) hypothesize that the presence of rebates and fees should not affect market equilibrium because bid-ask spreads would

⁴ Non-marketable limit orders are orders not immediately executable: a buy order priced lower than the best ask or a sell order priced higher than the best bid.

⁵ For more details on the experiment, see <https://www.sec.gov/news/press-release/2018-298>.

⁶ https://www.wsj.com/articles/stock-exchanges-to-test-trading-minus-rebates-seen-as-posing-conflict-11545256980?mod=article_inline&mod=article_inline

⁷ https://www.wsj.com/articles/nasdaq-joins-new-york-stock-exchange-in-suing-to-block-sec-rule-11550259637?mod=article_inline

automatically adjust and offset differences in fees/rebates. This is consistent with the work produced by Colliard and Foucault (2012)—their model suggests that only changes in the net trading costs should affect actual trading volume and market liquidity. Another angle is that, by attracting different investors, market impact differs across venues, which might ultimately compensate for the difference in fees. In sum, although it is clear that fees set by trading venues are an important element driving market equilibrium and that a better understanding of their effects is key to both investors and policy makers, empirical evidence on the impact of venue fees is still missing.

The main challenge in investigating these issues empirically is the lack of granular data on institutional investors' trading behavior and trading costs, making it challenging to properly assess the differences across venues. In this paper, we are able to address this challenge by using detailed trade data from a single large institutional investor with a consistent trading style (a flexible trading approach that avoids demanding immediacy). This allows us to compare transaction costs across venues ensuring that the results are not driven by systematic differences in trading approaches across investors or across time periods. While our findings might not apply to investors using alternative trading styles (e.g., immediacy-seeking market participants), we believe that our insights are applicable to other large institutional investors trading diversified portfolios.

In the main part of the analysis, we use trade data aggregated at the stock/day/side/venue type level to compare a few different measures of transaction costs across maker-taker and inverted exchanges: gross effective spread, net (of fees and rebates) effective spread, market impact, and net realized spread, which combines the net effective spread and the market impact. The main result is that, while market impact significantly differs across exchanges, with maker-taker exchanges exhibiting a two basis-point larger impact, the net realized spread does not differ across venues. This result is robust to the inclusion of a set of controls for stock and trade characteristics, as well as day and stock fixed effects.

We also investigate whether this baseline result varies across stock or market characteristics. For instance, while these results might hold in general, there might be subsets of stocks, e.g., low-priced or underperforming stocks, for which different venues can provide significantly different trading costs. Similarly, our results might not hold during periods of market turmoil if market impact reflects market conditions more promptly than fees and rebates. We do not find support for these hypotheses. In fact, there are no reliable differences in net realized spread based on a stock's price and past performance. Similarly, net realized spread is on average the same across market conditions, e.g., high- or low-volatility periods, although market impact does reliably differ across these periods.

Finally, we run the analysis at the individual trade level. This granular data set allows us to examine a much larger set of observations and expand the power of the tests, but requires us to adjust for potentially strong correlations across trades of each stock and across trades each day. To do that, we adjust standard errors by double clustering at the stock and day level. The results based on the more granular data set are consistent with the results from the aggregated one.

Overall, our findings are informative about the impact of different fee structures on the US market microstructure. Specifically, net trading costs seem to be independent of the choice of venue due to the offsetting relation between fees/rebates and market impact. The implication is that if commissions charged to investors are set to reflect the fees/rebates experienced by brokers, investors should be indifferent as to which type of venues brokers route their orders to. In other words, for investors who know where their orders are routed and have the ability to negotiate brokers' commissions, different exchange fee structures do not imply different overall costs.

2. Data

Our data comprise two main sets of information. First, we employ a proprietary in-house tick data set of market trades and quotes to compute transaction costs.⁸ The data cover about 3,000 securities across all US venues (exchanges, dark pools, off-exchange) with millisecond timestamp accuracy. Second, we obtain order executions from a large institutional investor (Dimensional Fund Advisors) that collects data from brokers on a daily basis. This database contains approximately 9 million observations for all trading days in 2017, with detailed information about the security identifier, price, size, venue, visibility (displayed/non-displayed), Time-In-Force (TIF)⁹, and timestamps of each order. Since fees and rebates are not explicitly provided in the order execution data, we estimate them using the Venue Fee Schedule available in **Table 1**. For our baseline specification, we assume mid-tier making fees and rebates. Later, we verify the robustness of our results by conducting the analysis using the highest- and lowest-tier making fees and rebates.

There are several advantages of this execution data set compared to standard trade and quote (TAQ) data sets. First, the side of the trade is known explicitly. We observe whether each trade is buyer- or seller-initiated, which is key to determine its associated exchange fee/rebate. This key information avoids the potential mislabeling of trades by using the Lee-Ready¹⁰ algorithm in market samples. Furthermore, since we use trades from the same market participant with a consistent trading style over time, our findings are not driven by heterogeneous trading behavior and different market access opportunities among investors.

Panel A of **Table 2** presents time-series statistics about the execution data set. The sample covers 250 trading days and 3,322 securities, with about 600 distinct securities traded daily on average. Panel B of **Table 2** reports cross-sectional descriptive statistics about the security universe. The traded stocks exhibit wide dispersion in terms of prices, market capitalizations, queue sizes, and bid-ask spreads. This will allow us to examine whether overall trading costs across venues vary with these stock characteristics.

⁸ The tick data set referred to here is compiled using intraday trade and quote data from Refinitiv Tick History (TRTH) and maintained by Dimensional Fund Advisors.

⁹ Time-In-Force is the amount of time an order will remain active before it is executed or expires. Common options include "Day" (cancel at close of the trading day), "Fill-or-Kill" (cancel if the entire order cannot be filled immediately), and "Immediate-or-Cancel" (cancel the part of order that is not filled immediately).

¹⁰ See Lee and Ready (1991).

Finally, Panels C–E provide summary statistics on the order side, price, and venue of the data set. We see that the proprietary trading data set is broadly diversified across maker-taker and inverted exchanges, which can allow us to draw robust comparisons of trading costs across venues.

3. Methodology

This section explains how we compute some of the key variables used in the analysis and our approach in estimating the differences in those variables between maker-taker and inverted types of exchanges.

The net realized spread is the main outcome variable and is defined as the sum of the net effective spread and the market impact. The net effective spread is computed as the sum of the gross effective spread and the rebate/fee, where the gross effective spread is computed as follows:

$$\text{Gross Effective Spread} = D_{it} \frac{(V_{it} - P_{it})}{V_{it}}$$

while the rebate/fee is simply given by the following:

$$\text{Rebate or Fee} = -R_i/V_{it}$$

where P_{it} is the transaction price for security i at time t , D_{it} is an indicator variable that is +1 for the liquidity provider's buy trades (or seller-initiated trades), and -1 for the liquidity provider's sell trades (or buyer-initiated trades), R_i is the estimated rebate (negative) or fee (positive) from Table 1, and V_{it} is the quoted national best bid and offer (NBBO) midpoint at time t . The signs of the variables are chosen so that the more negative the net effective spread, the higher the trading costs.

Next, we compute market impact:¹¹

$$\text{Market Impact} = D_{it} \frac{(V_{it+n} - V_{it})}{V_{it}}$$

V_{it+n} is the quoted NBBO midpoint at time $t+n$. In our baseline specification, we compute the market impact over the five-minute window after the trade, i.e., $n=5$.

Finally, the net realized spread is given by:

$$\text{Net Realized Spread} = \text{Net Effective Spread} + \text{Market Impact} = \frac{D_{it}(V_{it+n} - P_{it}) - R_i}{V_{it}}$$

Table 3 compares average trading costs across maker-taker and inverted exchanges by reporting the average differences in spreads, fees, and market impact. For each stock-day observation for which individual trades are found in both maker-taker and inverted exchanges,¹² we compute trade-value weighted average costs in maker-taker and inverted exchanges separately and calculate the difference. We then take the trade-value weighted average across all stocks for each day. These

¹¹ This approach is used in academic studies such as Bessembinder and Venkataraman (2010).

¹² 9.7% of individual trades are dropped in terms of trading value because those stocks are traded in only one exchange type on the day.

trade-value weighted daily differences are used in a trade-value weighted t-test. Already these simple statistics show that while there is a difference in market impact between the two types of venues, this difference is less pronounced or absent once fees are considered.

To study further the impact of fee structure on market impact and net realized spread, we apply the following baseline specification to trades that are aggregated¹³ by stock, day, side, and exchange type:

$$\text{Market Impact}_{it} = \beta^{MI} * [\text{Maker} - \text{Taker}] + \boldsymbol{\Omega}^{MI} * \text{Controls} + (\gamma_i^{MI}) + (\gamma_t^{MI}) + \varepsilon_{it}^{MI}$$

and

$$\text{Net Realized Spread}_{it} = \beta^{RS} * [\text{Maker} - \text{Taker}] + \boldsymbol{\Omega}^{RS} * \text{Controls} + (\gamma_i^{RS}) + (\gamma_t^{RS}) + \varepsilon_{it}^{RS}$$

Maker-Taker is a dummy variable identifying the type of exchange (1 for maker-taker and 0 for inverted), and β is the main coefficient of interest. To absorb variation across trades, we add a vector of “Controls,” which includes some potentially important characteristics, such as the side of the trade, the bid-ask spread, and the order participant rate in the market. To allow for arbitrary correlation within the same stock or day cluster, we adjust standard errors by double clustering at the stock and day level. To control in a non-parametric way for differences among trades, in the most conservative specifications, we include both stock and day fixed effects γ_i and γ_t . Later, we also run the model using individual-level trades (without aggregation), double clustering the standard errors at the stock and day level and including time-window fixed effects.

4. Main Results

We start by analyzing whether market impact differs across exchanges that offer a maker-taker fee structure and exchanges with an inverted structure. Columns 1–5 in Panel A of **Table 4** include specifications with different control variables. To absorb time-invariant stock characteristics, we include stock fixed effects in Column 6, while Column 7 presents the results including day fixed effects that are helpful in absorbing market trends. The most conservative specification is presented in Column 8, where we include both stock and day fixed effects. In all specifications, we consistently find that the maker-taker exchanges exhibit larger market impact by about two basis points, i.e., the coefficient on the exchange type dummy is statistically reliable at the 1% level. The relative stability of the coefficient across specifications suggests that differences in market impact across venues are not correlated with the vector of controls or with systematic differences across time or stocks.

At first glance, these results would seem to suggest that the venues’ fee structure significantly affects transaction costs, with higher transaction costs associated with trading on maker-taker exchanges. However, the previous results do not consider other important components of transaction costs: fees and spreads. In Panel B of Table 4, we replace market impact with net realized spread as the dependent variable to test whether the changes in market impact are compensated by the differences

¹³ The reason we aggregate trades by stock, day, side, and exchange type is to avoid potential issues with serial correlation within clusters of trades.

in fees and spreads. That is, maker-taker exchanges could charge different fees and offer rebates that would end up compensating investors for the differences in market impact we observed in Panel A. Similarly, differences in market impact could be offset by differences in prices relative to the midpoint between the bid and the ask at time of execution.

Panel B shows that net realized spreads are not reliably different across exchanges. The main coefficient of interest is not reliably different from zero and is also economically small. This conclusion holds across all model specifications, regardless of the controls included, suggesting that any difference in market impact experienced by investors on a given exchange tends to be offset by the difference in rebates/fees. Effectively, after accounting for fees and spreads, trading costs appear to be similar between different types of exchanges.

With respect to the observed relations between the dependent variables and the control variables, the results are intuitive. We find costs to be higher for larger-spread and lower-priced stocks. There is also a tradeoff between speed and cost, as captured by the negative and reliable coefficient for the parent order participant rate in the market. This is expected, since the faster and more aggressive trades are in the market, the higher the costs are likely to be. Finally, on average, costs appear unrelated to the side of the trade.

It is natural to wonder whether our main results vary based on market conditions. For instance, periods of high uncertainty might exhibit significantly higher market impact, which makes the venue choice more important, particularly if fees and rebates compensate for differences in market impact on average but not in more extreme market conditions. We test this hypothesis in the remaining Panels of Table 4 by estimating the baseline specification by subsample. In Panels C and D of Table 4, we split the sample by the previous day's market volatility. Panel C focuses on implied volatility, using the CBOE Volatility Index (VIX) as our metric of choice. Panel D uses realized volatility¹⁴ of the S&P 500 Index over the last 20 trading days. While in Panel C we find reliably negative coefficients for maker-taker venues on days with lower market volatility, in Panel D none of the coefficients on the main variable of interest are reliably different from zero. In summary, the net realized spread does not appear to differ reliably across venue types in periods of high or low market volatility.

We also examine whether our main results vary with stock characteristics. Panel E reports the estimates separately for low-priced (Columns 1–4) and high-priced stocks (Columns 5–8), where stock price is the closing price for the same trading day. Intuitively, the costs of low-priced stocks are more impacted by choice of exchange, because exchange fee/rebate, which is paid in cents-per-share, increases in basis points as stock price decreases. However, we find that the coefficients on the main variable of interest, the exchange identifier dummy, are never reliably different from zero. Panel F tests whether a stock's past performance affects investors' transaction costs and reports the results separately based on the previous 20 trading days' cumulative returns, differentiating between negative (Columns 1–4) and positive (Columns 5–8) past returns. Intuitively, transaction costs for

¹⁴ Realized volatility is computed as the annualized standard deviation of daily returns, using a 20-day rolling window.

stocks experiencing large directional returns might be higher if market makers are less willing to provide liquidity during those periods. Consistently across subsamples, we find instead that the net realized spread does not differ based upon the type of exchange, e.g., the coefficient on the maker-taker dummy is never reliably different from zero.

One potential concern with the previous analysis is that we do not observe trade-by-trade fees and rebates;¹⁵ rather, we estimate them using mid-tier rates. To check whether this might be a concern, we compute the net realized spread using fees and rebates from the highest and lowest tiers of the fee schedule. Intuitively, it could be that on average transaction costs are the same for the average investor, but there might be significant heterogeneity depending on the fee tier. Panels A and B of **Table 5** present the results with the highest- and lowest-tier fee/rebate, respectively. Even with these alternative approaches to computing our outcome variable, we find that the net realized spread does not differ reliably based on the fee structure of the exchange. This finding supports the robustness of our main results.

Finally, we repeat our main analysis at the most granular level possible—the individual trade level. This allows us to compare across venues the transaction costs of individual trades rather than the transaction costs aggregated at the stock-day-side-venue type level. To account for the potentially high correlation across trades within the same stock or day cluster, we adjust standard errors by double-clustering at the stock and day level. Panel A of **Table 6** reports the results for market impact, while Panel B reports the results for net realized spread. Column 1 includes time-window fixed effects (30-minute windows starting from 12:00 am) and Column 2 also includes stock fixed effects, while Columns 3 and 4 control for day fixed effects. Similar to the analysis on the aggregated data, we find that maker-taker exchanges exhibit about 2 basis points larger market impact than inverted venues. We find a similar tradeoff between speed and cost, with larger market impact for larger orders, as measured by the fraction of the daily volume, and for orders with larger spreads. Panel B complements the previous analysis by showing that net realized spreads are indistinguishable across exchanges, as the main coefficient of interest is small and unreliable in most specifications.

5. Conclusion

Investors today have a wide variety of exchanges and off-market venues to select from when deciding where to trade. The venues differ in a variety of ways, including by transaction cost. The quoted bid-offer is one component of that transaction cost and is readily observable. However, it does not tell the full story, and total trading costs can differ materially after factoring in market impact, fees, and rebates.

To gain a more comprehensive understanding of the total trading costs facing investors, we analyze a granular data set of trades from a large institutional investor. We find that market impact does

¹⁵ For exchanges where fees and rebates are determined by volume-based sliding scales, the actual fees and rebates typically are determined at the end of a calendar month based on a member's transaction volume during the month. Without knowing a broker's total transaction volumes, we are not able to infer the actual fee and rebate for each trade.

indeed differ across venues, with maker-taker venues exhibiting significantly larger market impact. However, after combining market impact with net effective spread, we find that the net realized spreads—the total implicit trading costs faced by investors—are not reliably different across exchange types. Further, we find that this conclusion holds across a variety of market conditions and stock characteristics.

For institutional investors capable of negotiating commissions so that they are in line with the fees/rebates experienced by their brokers, our work suggests that pro-rata allocation across venues is the optimal approach, given no meaningful difference in transaction costs across exchange types. However, it's worth noting that conclusions may differ for retail investors. At times, the current market equilibrium may result in suboptimal venue selection for this segment of the investment universe, as retail investors may not experience net trading costs in the same way. Retail investors are unlikely to be able to negotiate commissions in line with the fees/rebates experienced by their brokers. As a result, it is possible that their trades get routed to the venues with the largest rebates and market impact, but only the market impact gets reflected in the trading costs that retail investors experience. This may be a sensible area for additional attention from regulators, who could consider requiring the pass-through of all rebates/fees. Improving the transparency of total trading costs could also be an area for greater regulatory focus.

Tables

TABLE 1

Summary of Venues

This table gives the trading fee schedule and total traded value for all US equities exchanges in 2017. For fee schedule, each number in the table represents the dollar amount of the rebate (with parentheses) or fee (without parentheses) for every 1 share traded.

MIC	Venue Type	Venue Full Name	Fee Schedule					Traded Value ¹⁶	
			Make Display (Mid-Tier)	Make Display (Lowest Tier)	Make Display (Highest Tier)	Make Non-Display	Take (Mid-Tier)	Dollar Value Traded (2017)	% of Total Value Traded (2017)
ARCX	Maker-Taker	NYSE Arca	(\$0.0024)	(\$0.0024)	(\$0.0030)	\$0.0000	\$0.0030	\$3,475.5 B	10.6%
BATS	Maker-Taker	BATS	(\$0.0022)	(\$0.0015)	(\$0.0030)	(\$0.0015)	\$0.0030	\$2,825.2 B	8.6%
BATY	Inverted	BATY	\$0.0019	\$0.0024	\$0.0012	\$0.0024	(\$0.0010)	\$1,930.1 B	5.9%
EDGA ¹⁷	Inverted	EDGA	\$0.0022	\$0.0030	\$0.0016	\$0.0030	(\$0.0024)	\$712.8 B	2.2%
EDGX	Maker-Taker	EDGX	(\$0.0020)	(\$0.0017)	(\$0.0029)	(\$0.0015)	\$0.0030	\$3,006.2 B	9.2%
IEXG	Flat-Fee	IEX	\$0	\$0	\$0	\$0.0009	\$0.0005	\$1,333.0 B	4.1%
XASE	Flat-Fee	NYSE American	\$0.0000	\$0.0000	\$0.0000	\$0.0002	\$0.0002	\$41.4 B	0.1%
XBOS	Inverted	NASDAQ BX	\$0.0018	\$0.0020	\$0.0014	\$0.0024	(\$0.0016)	\$1,610.3 B	4.9%
XCHI	Maker-Taker	NYSE Chicago	(\$0.0020)	(\$0.0020)	(\$0.0020)	\$0.0000	\$0.0030	\$180.6 B	0.6%
XCIS	Inverted	NYSE National	\$0.0023	\$0.0023	\$0.0010	\$0.0025	(\$0.0020)	\$5 B	0.0%
XNAS	Maker-Taker	NASDAQ	(\$0.0025)	(\$0.0020)	(\$0.00305)	(\$0.0008)	\$0.0030	\$9,272.4 B	28.3%
XNYS	Maker-Taker	NYSE	(\$0.0019)	(\$0.0012)	(\$0.0022)	(\$0.0012)	\$0.0028	\$8,011.8 B	24.5%
XPSX	Maker-Taker	NASDAQ PSX	(\$0.0028)	(\$0.0023)	(\$0.0030)	\$0	\$0.0030	\$329.4 B	1.0%

¹⁶ Source: Dimensional using Refinitiv data. In USD. Includes only ordinary shares and ADRs (American depositary receipt), excludes ETPs (exchange-traded products). From 2017/01/01 to 2017/12/31.

¹⁷ We excluded EDGA in the analysis because its fee schedule had a major change from inverted model to flat-fee model on June 1, 2017. See https://cdn.cboe.com/resources/fee_schedule/2017/Updated-Bats-Equity-Exchange-Fee-Schedule-Update-Effective-June-1-2017.pdf.

TABLE 2

Descriptive Statistics

The tables below provide metrics describing the order execution data set from Dimensional Fund Advisors used in the analysis. Panel A provides time-series statistics of the stock universe covered by the data set. Panel B provides cross-sectional statistics of the stock universe. For each security, we compute the median of daily metrics in 2017, including market capitalization, stock price, daily average queue size, and daily average bid-ask spread, and then report the cross-sectional descriptive statistics in Panel B. Market capitalization (MCAP) is defined as shares outstanding multiplied by the stock's closing price. Price is defined as the stock's closing prices. Queue size is the average of National Best Bid Size and National Best Ask Size. Spread is the bid-ask spread between National Best Bid Price and National Best Ask Price. Panels C to E summarize the characteristics of the order execution data set. Order side and pricing are described in Panels C and D, and distribution of orders/trades by venue are described in Panel E.

PANEL A: NUMBER OF TRADING DAYS AND TRADED STOCKS

Variable	Count
Trading Days	250
Distinct Traded Securities	3,322
Average Distinct Securities Traded Daily	578
25th Percentile Distinct Securities Traded Daily	470
Median Distinct Securities Traded Daily	565
75th Percentile Distinct Securities Traded Daily	670

PANEL B: CHARACTERISTICS OF TRADED STOCKS

Variable	Average	25th	Median	75th
MCAP (MM USD)	10,090	417	1,545	5,545
Price (USD)	47	13	28	55
Queue Size (shares)	4,663	426	839	2,299
Spread (bps)	43	7	18	48

PANEL C: ORDER SIDE

Side	Count	% of Orders
Buy	5,122,021	71.00%
Sell	2,091,901	29.00%

PANEL D: ORDER PRICING

Order Pricing ¹⁸	Count	% of Orders
At-Quote	6,112,448	84.73%
Behind-Quote	1,059,157	14.68%
Inside-Quote	42,317	0.59%

PANEL E: ORDER VENUE

Venue Type	Order Count	Trade Count	Traded Value (USD)	% of Traded Value
Maker-Taker	4,701,910	5,978,135	17,971,545,426	72.46%
Inverted	2,465,520	3,296,776	6,832,113,709	27.54%

¹⁸ Order pricing is determined by comparing order prices to national best bid/ask prices at the time when an order is placed. "At-Quote" means buy on best bid or sell on best ask, "Behind-Quote" means buy below best bid or sell above best ask, and "Inside-Quote" means buy above best bid or sell below best ask.

TABLE 3

Trading Costs in Maker-Taker and Inverted Exchanges

We break down trading costs into gross effective spread, fee/rebate, net effective spread, market impact, and net realized spread (in basis points) and compare each component between maker-taker and inverted exchanges. The differences are calculated by taking the difference in trade-value weighted average costs for maker-taker and inverted exchanges for each stock and day. Then we take trade-value weighted average across stocks for each day. These trade-value weighted daily differences are used in a weighted t-test. Computations of costs in maker-taker and inverted exchanges follow the same methodology.

	Gross Effective Spread	Fee or Rebate	Net Effective Spread	Market Impact	Net Realized Spread
Exchange Type	A	B	C = A + B	D	E = C + D
Maker-Taker	4.31	0.89	5.20	-5.58	-0.38
Inverted	4.30	-0.76	3.53	-3.87	-0.34
difference	0.02	1.65	1.67	-1.71	-0.04
t-statistic	1.90	14.80	14.80	-8.60	-0.20

TABLE 4

Market Impact and Net Realized Spread using Aggregated-Level Data

We build multivariable linear regression models to examine the relation, in basis points, between exchange type and market impact (Panel A), as well as exchange type and net realized spread (Panel B). Trades are aggregated by stock, day, side, and exchange type before fitting the regression model. Independent variables are as follows. *ExchangeType* is a categorical variable with a value of 1 for “maker-taker” and 0 for “inverted.” *ParentOrder_POV* is the trade-value weighted average parent order participant rate over market consolidated volume during the time when the parent order is exposed. *StockSpreadInBps* is the median of a stock’s daily time-weighted average bid-ask spreads in basis points in 2017. *LogStockPrice* is the natural logarithm of a stock’s average closing price in 2017. *LogStockMCAPFF* is the natural logarithm of a stock’s average free-float adjusted market capitalization in USD in 2017. *Side* is a categorical variable with a value of 1 for sell and 0 for buy. We also introduce two types of fixed effects – Stock FE (on stock level) and Day FE (on trade date level). We split the sample by previous day’s market implied volatility (VIX) (Panel C), realized volatility of S&P 500 (Panel D), stock’s closing price (Panel E) and stock’s previous 20-day cumulative return (Panel F). Standard errors are shown below the coefficient estimates. Standard errors are clustered by stock and trade date.

PANEL A: MARKET IMPACT

Dependent Variable Is Market Impact (in bps) Baseline Specification								
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
ExchangeType	-1.338*** (0.087)	-1.452*** (0.066)	-1.779*** (0.078)	-1.761*** (0.080)	-1.741*** (0.079)	-1.791*** (0.079)	-1.740*** (0.080)	-1.790*** (0.080)
ParentOrder_POV	-85.595*** (5.736)	-8.298*** (2.641)	-8.068*** (2.666)	-6.707** (2.708)	-6.451** (2.710)	-13.807*** (3.000)	-8.299*** (2.649)	-16.248*** (2.911)
StockSpreadInBps		-0.360*** (0.006)	-0.334*** (0.005)	-0.331*** (0.006)	-0.331*** (0.006)		-0.331*** (0.006)	
LogStockPrice			1.276*** (0.073)	1.212*** (0.088)	1.236*** (0.090)		1.215*** (0.090)	
LogStockMCAPFF				0.065 (0.043)	0.063 (0.044)		0.073 (0.045)	
Side					-0.348** (0.163)	-0.357* (0.197)	-0.261 (0.162)	-0.255 (0.193)
Constant	-1.466*** (0.248)	-0.019 (0.106)	-4.731*** (0.292)	-6.056*** (0.902)	-6.010*** (0.920)			
Stock FE						Yes	Yes	
Day FE							Yes	Yes
SE CL	Stock and Day	Stock and Day	Stock and Day	Stock and Day	Stock and Day	Stock and Day	Stock and Day	Stock and Day
Observations	279,071	279,071	279,071	279,071	279,071	279,071	279,071	279,071
R ²	0.026	0.248	0.257	0.257	0.257	0.295	0.263	0.302
Adjusted R ²	0.026	0.248	0.257	0.257	0.257	0.288	0.263	0.294

Note:

* $p < 0.1$; ** $p < 0.05$; *** $p < 0.01$

PANEL B: NET REALIZED SPREAD

Dependent Variable Is Net Realized Spread (in bps) Baseline Specification								
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
ExchangeType	0.009 (0.054)	-0.0001 (0.054)	-0.085 (0.057)	-0.077 (0.056)	-0.062 (0.056)	-0.077 (0.055)	-0.060 (0.056)	-0.074 (0.055)
ParentOrder_POV	-20.277*** (2.432)	-14.096*** (2.382)	-14.036*** (2.434)	-13.388*** (2.542)	-13.193*** (2.551)	-12.724*** (2.754)	-15.120*** (2.417)	-14.748*** (2.643)
StockSpreadInBps		-0.029*** (0.004)	-0.022*** (0.004)	-0.020*** (0.005)	-0.021*** (0.005)		-0.020*** (0.005)	
LogStockPrice			0.333*** (0.055)	0.302*** (0.060)	0.321*** (0.060)		0.304*** (0.060)	
LogStockMCAFF				0.031 (0.032)	0.030 (0.033)		0.035 (0.034)	
Side					-0.265* (0.153)	-0.250 (0.185)	-0.223 (0.154)	-0.213 (0.185)
Constant	0.021 (0.099)	0.137 (0.097)	-1.092*** (0.234)	-1.723** (0.726)	-1.688** (0.744)			
Stock FE						Yes		Yes
Day FE							Yes	Yes
SE CL	Stock and Day	Stock and Day	Stock and Day	Stock and Day	Stock and Day	Stock and Day	Stock and Day	Stock and Day
Observations	279,071	279,071	279,071	279,071	279,071	279,071	279,071	279,071
R ²	0.002	0.004	0.005	0.005	0.005	0.041	0.013	0.048
Adjusted R ²	0.002	0.004	0.005	0.005	0.005	0.030	0.012	0.037

Note:

*p<0.1; **p<0.05; ***p<0.01

PANEL C: SAMPLE SPLIT BY IMPLIED MARKET VOLATILITY (VIX)

Dependent Variable Is Net Realized Spread (in bps) Sample Split by Above/Below Median Implied Market Volatility (VIX)								
	Low (1)	Low (2)	Low (3)	Low (4)	High (5)	High (6)	High (7)	High (8)
ExchangeType	-0.107* (0.061)	-0.117** (0.057)	-0.107* (0.061)	-0.118** (0.058)	-0.019 (0.081)	-0.020 (0.080)	-0.017 (0.081)	-0.015 (0.080)
ParentOrder_POV	-13.978*** (3.611)	-13.344*** (3.539)	-16.882*** (3.222)	-16.097*** (3.317)	-12.455*** (3.424)	-13.435*** (3.726)	-13.458*** (3.448)	-14.185*** (3.666)
StockSpreadlnBps	-0.018*** (0.006)		-0.018*** (0.006)		-0.024*** (0.007)		-0.023*** (0.007)	
LogStockPrice	0.393*** (0.084)		0.384*** (0.084)		0.244*** (0.087)		0.230*** (0.085)	
LogStockMCAPFF	0.020 (0.045)		0.019 (0.045)		0.040 (0.051)		0.049 (0.049)	
Side	-0.103 (0.169)	-0.175 (0.222)	-0.065 (0.164)	-0.132 (0.223)	-0.415 (0.252)	-0.416 (0.291)	-0.369 (0.253)	-0.368 (0.295)
Constant	-1.686 (1.038)				-1.696 (1.125)			
Stock FE	Yes		Yes		Yes		Yes	
Day FE			Yes				Yes	
SE CL	Stock and Day	Stock and Day	Stock and Day	Stock and Day	Stock and Day	Stock and Day	Stock and Day	Stock and Day
Observations	141,213	141,213	141,213	141,213	137,858	137,858	137,858	137,858
R ²	0.006	0.062	0.013	0.069	0.005	0.059	0.012	0.066
Adjusted R ²	0.006	0.043	0.012	0.049	0.005	0.039	0.011	0.045

Note:

*p<0.1; **p<0.05; ***p<0.01

PANEL D: SAMPLE SPLIT BY REALIZED MARKET VOLATILITY (RV)

Dependent Variable Is Net Realized Spread (in bps) Sample Split by Above/Below Median Realized Market Volatility (RV)								
	Low (1)	Low (2)	Low (3)	Low (4)	High (5)	High (6)	High (7)	High (8)
ExchangeType	-0.027 (0.066)	-0.026 (0.066)	-0.035 (0.069)	-0.037 (0.068)	-0.094 (0.073)	-0.112 (0.070)	-0.084 (0.071)	-0.102 (0.068)
ParentOrder_POV	-12.189*** (3.768)	-11.135** (4.266)	-13.245*** (3.695)	-12.211*** (4.141)	-14.280*** (3.380)	-12.742*** (3.449)	-17.073*** (2.913)	-15.850*** (3.038)
StockSpreadlnBps	-0.019*** (0.006)		-0.019*** (0.006)		-0.022*** (0.007)		-0.022*** (0.007)	
LogStockPrice	0.274*** (0.078)		0.269*** (0.076)		0.371*** (0.084)		0.344*** (0.085)	
LogStockMCAPFF	0.037 (0.041)		0.039 (0.043)		0.023 (0.047)		0.029 (0.048)	
Side	-0.272 (0.230)	-0.211 (0.284)	-0.137 (0.234)	-0.044 (0.289)	-0.258 (0.205)	-0.331 (0.248)	-0.304 (0.201)	-0.428* (0.239)
Constant	-1.737* (0.921)				-1.662 (1.071)			
Stock FE		Yes		Yes		Yes		Yes
Day FE			Yes	Yes			Yes	Yes
SE CL	Stock and Day	Stock and Day	Stock and Day	Stock and Day	Stock and Day	Stock and Day	Stock and Day	Stock and Day
Observations	132,603	132,603	132,603	132,603	146,468	146,468	146,468	146,468
R ²	0.005	0.055	0.011	0.061	0.006	0.061	0.014	0.070
Adjusted R ²	0.005	0.034	0.010	0.040	0.006	0.043	0.013	0.050

Note:

*p<0.1; **p<0.05; ***p<0.01

PANEL E: SAMPLE SPLIT BY STOCK PRICE

Dependent Variable Is Net Realized Spread (in bps) Sample Split by Stock Closing Price (Less or Greater than \$20)								
	Low (1)	Low (2)	Low (3)	Low (4)	High (5)	High (6)	High (7)	High (8)
ExchangeType	-0.133 (0.147)	-0.108 (0.144)	-0.172 (0.147)	-0.141 (0.143)	-0.042 (0.048)	-0.067 (0.049)	-0.027 (0.049)	-0.052 (0.049)
ParentOrder_POV	-16.267*** (4.819)	-15.669*** (5.289)	-19.756*** (4.977)	-20.180*** (5.364)	-11.289*** (2.376)	-11.353*** (2.540)	-13.022*** (2.176)	-12.948*** (2.390)
StockSpreadInBps	-0.002 (0.008)		-0.002 (0.008)		-0.025*** (0.006)		-0.024*** (0.006)	
LogStockPrice	0.867*** (0.175)		0.893*** (0.179)		0.140* (0.079)		0.109 (0.076)	
LogStockMCAPIF	0.353*** (0.111)		0.345*** (0.115)		0.005 (0.034)		0.014 (0.034)	
Side	-0.242 (0.282)	-0.169 (0.366)	-0.135 (0.283)	-0.116 (0.357)	-0.277* (0.142)	-0.267 (0.171)	-0.258* (0.142)	-0.237 (0.170)
Constant	-10.201*** (2.420)				-0.394 (0.731)			
Stock FE		Yes		Yes		Yes		Yes
Day FE			Yes	Yes			Yes	Yes
SE CL	Stock and Day	Stock and Day	Stock and Day	Stock and Day	Stock and Day	Stock and Day	Stock and Day	Stock and Day
Observations	121,868	121,868	121,868	121,868	157,203	157,203	157,203	157,203
R ²	0.004	0.050	0.017	0.063	0.004	0.033	0.014	0.043
Adjusted R ²	0.004	0.040	0.015	0.051	0.004	0.020	0.012	0.029

Note:

*p<0.1; **p<0.05; ***p<0.01

PANEL F: SAMPLE SPLIT BY PAST RETURNS

Dependent Variable Is Net Realized Spread (in bps) Sample Split by Positive/Negative Cumulative Returns Over Previous 20 Days								
	Negative (1)	Negative (2)	Negative (3)	Negative (4)	Positive (5)	Positive (6)	Positive (7)	Positive (8)
ExchangeType	-0.045 (0.081)	-0.057 (0.080)	-0.044 (0.080)	-0.060 (0.079)	-0.073 (0.057)	-0.091 (0.056)	-0.071 (0.057)	-0.084 (0.056)
ParentOrder_POV	-16.305*** (3.369)	-15.297*** (3.678)	-18.601*** (3.446)	-17.578*** (3.654)	-10.611*** (2.983)	-10.742*** (3.090)	-12.475*** (2.607)	-13.005*** (2.772)
StockSpreadInBps	-0.017** (0.007)		-0.016** (0.007)		-0.024*** (0.005)		-0.024*** (0.005)	
LogStockPrice	0.380*** (0.085)		0.353*** (0.089)		0.279*** (0.068)		0.277*** (0.065)	
LogStockMCAPIF	0.003 (0.042)		0.017 (0.045)		0.049 (0.038)		0.045 (0.038)	
Side	-0.421* (0.224)	-0.423 (0.266)	-0.377 (0.238)	-0.364 (0.270)	-0.150 (0.143)	-0.195 (0.186)	-0.110 (0.134)	-0.147 (0.179)
Constant	-1.184 (0.964)				-2.095** (0.829)			
Stock FE	Yes		Yes		Yes		Yes	
Day FE			Yes				Yes	
SE CL	Stock and Day	Stock and Day	Stock and Day	Stock and Day	Stock and Day	Stock and Day	Stock and Day	Stock and Day
Observations	127,127	127,127	127,127	127,127	151,920	151,920	151,920	151,920
R ²	0.005	0.061	0.017	0.073	0.006	0.062	0.015	0.071
Adjusted R ²	0.005	0.039	0.015	0.050	0.006	0.044	0.013	0.051

Note:

*p<0.1; **p<0.05; ***p<0.01

TABLE 5

Robustness to Fee/Rebate Determination

We build multivariable linear regression models to examine the relation, in basis points, between exchange type and net realized spread. Trades are aggregated by stock, day, side, and exchange type before fitting the regression model. Panel A uses the highest-tier fee/rebate, while Panel B uses the lowest tier. Independent variables are as follows. *ExchangeType* is a categorical variable with a value of 1 for “maker-taker” and 0 for “inverted.” *ParentOrder_POV* is the trade-value weighted average parent order participant rate over market consolidated volume during the time when the parent order is exposed. *StockSpreadInBps* is the median of a stock’s daily time-weighted average bid-ask spreads in basis points in 2017. *LogStockPrice* is the natural logarithm of a stock’s average closing price in 2017. *LogStockMCAPFF* is the natural logarithm of a stock’s average free-float adjusted market capitalization in USD in 2017. *Side* is a categorical variable with a value of 1 for sell and 0 for buy. We also introduce two types of fixed effects – Stock FE (on stock level) and Day FE (on trade date level). Standard errors are shown below the coefficient estimates. Standard errors are clustered by stock and trade date.

PANEL A: USING THE HIGHEST-TIER FEE/REBATE

Dependent Variable Is Net Realized Spread (in bps) Assuming Highest Fee Tier								
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
ExchangeType	-0.046 (0.053)	-0.052 (0.054)	-0.075 (0.056)	-0.060 (0.056)	-0.046 (0.056)	-0.067 (0.055)	-0.045 (0.056)	-0.065 (0.055)
ParentOrder_POV	-19.233*** (2.429)	-14.958*** (2.445)	-14.942*** (2.461)	-13.770*** (2.549)	-13.590*** (2.557)	-13.004*** (2.764)	-15.533*** (2.417)	-15.033*** (2.653)
StockSpreadInBps		-0.020*** (0.004)	-0.018*** (0.004)	-0.015*** (0.005)	-0.015*** (0.005)		-0.015*** (0.005)	
LogStockPrice			0.090* (0.054)	0.035 (0.059)	0.052 (0.059)		0.033 (0.059)	
LogStockMCAPFF				0.056* (0.031)	0.054* (0.032)		0.059* (0.033)	
Side					-0.245 (0.152)	-0.245 (0.184)	-0.202 (0.154)	-0.209 (0.185)
Constant	0.271*** (0.097)	0.351*** (0.097)	0.019 (0.231)	-1.123 (0.706)	-1.090 (0.724)			
Stock FE						Yes	Yes	
Day FE							Yes	Yes
SE CL	Stock and Day	Stock and Day	Stock and Day	Stock and Day	Stock and Day	Stock and Day	Stock and Day	Stock and Day
Observations	279,071	279,071	279,071	279,071	279,071	279,071	279,071	279,071
R ²	0.002	0.003	0.003	0.003	0.003	0.038	0.010	0.045
Adjusted R ²	0.002	0.003	0.003	0.003	0.003	0.027	0.010	0.034

Note:

* $p < 0.1$; ** $p < 0.05$; *** $p < 0.01$

PANEL B: USING THE LOWEST-TIER FEE/REBATE

Dependent Variable Is Net Realized Spread (in bps) Assuming Lowest Fee Tier								
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
ExchangeType	0.040 (0.055)	0.029 (0.055)	-0.099* (0.057)	-0.096* (0.056)	-0.080 (0.057)	-0.090 (0.056)	-0.077 (0.057)	-0.087 (0.055)
ParentOrder_POV	-21.030*** (2.456)	-13.556*** (2.361)	-13.466*** (2.427)	-13.195*** (2.547)	-12.993*** (2.556)	-12.530*** (2.752)	-14.919*** (2.427)	-14.552*** (2.642)
StockSpreadInBps		-0.035*** (0.004)	-0.025*** (0.004)	-0.024*** (0.005)	-0.024*** (0.005)		-0.024*** (0.005)	
LogStockPrice			0.500*** (0.057)	0.487*** (0.063)	0.506*** (0.063)		0.491*** (0.063)	
LogStockMCAPFF				0.013 (0.034)	0.012 (0.035)		0.017 (0.035)	
Side					-0.275* (0.153)	-0.252 (0.185)	-0.235 (0.155)	-0.215 (0.185)
Constant	-0.145 (0.102)	-0.005 (0.099)	-1.851*** (0.241)	-2.115*** (0.753)	-2.078*** (0.771)			
Stock FE						Yes		Yes
Day FE							Yes	Yes
SE CL	Stock and Day	Stock and Day	Stock and Day	Stock and Day	Stock and Day	Stock and Day	Stock and Day	Stock and Day
Observations	279,071	279,071	279,071	279,071	279,071	279,071	279,071	279,071
R ²	0.002	0.005	0.007	0.007	0.007	0.044	0.015	0.051
Adjusted R ²	0.002	0.005	0.007	0.007	0.007	0.034	0.014	0.040

Note:

*p<0.1; **p<0.05; ***p<0.01

TABLE 6

Market Impact and Net Realized Spread with Trade-Level data

We build multivariable linear regression models to examine the relation between exchange type and market impact (Panel A) and net realized spread (Panel B) in basis points using individual trades by a large institutional investor. Independent variables include the following. *ExchangeType* is a categorical variable with a value of 1 for “maker-taker” and 0 for “inverted.” *ParentOrderPOV* is parent order participant rate over market consolidated volume during the time when parent order is exposed. *PctVolumeOverDailyTotal* is the ratio between trade volume and market daily total volume. *BidAskSpreadInBps* is a stock’s bid-ask spread in basis points at the time of execution. *LogTradePrice* is the natural logarithm of execution price. *LogTradeValue* is the natural logarithm of execution size in USD. *TradeSide* is a categorical variable with a value of 1 for sell and 0 for buy. We introduce three types of fixed effects – *TimeWindow FE* (30-minute windows starting from 12:00am), *Stock FE* (on stock level), and *Day FE* (on trade date level). Standard errors are shown below the coefficient estimates. Standard errors are clustered by stock and trade date.

PANEL A: MARKET IMPACT

Dependent Variable Is Market Impact (in bps) Using Individual Trades				
	(1)	(2)	(3)	(4)
ExchangeType	-2.139*** (0.078)	-1.840*** (0.074)	-2.128*** (0.079)	-1.832*** (0.075)
ParentOrderPOV	-17.908*** (2.232)	-8.072*** (1.834)	-19.467*** (2.157)	-9.796*** (1.765)
PctVolumeOverDailyTotal	-413.751*** (43.107)	-396.420*** (43.496)	-410.910*** (42.967)	-395.269*** (43.550)
BidAskSpreadInBps	-0.352*** (0.006)	-0.214*** (0.006)	-0.351*** (0.006)	-0.212*** (0.006)
LogTradePrice	2.125*** (0.142)	3.193*** (0.423)	2.117*** (0.142)	3.236*** (0.404)
LogTradeValue	-0.958*** (0.119)	-1.205*** (0.082)	-0.972*** (0.116)	-1.218*** (0.079)
TradeSide	-0.312* (0.171)	-0.448** (0.197)	-0.247 (0.171)	-0.356* (0.196)
TimeWindow FE	Yes	Yes	Yes	Yes
Stock FE		Yes		Yes
Day FE			Yes	Yes
SE CL	Stock and Day	Stock and Day	Stock and Day	Stock and Day
Observations	8,859,723	8,859,723	8,859,723	8,859,723
R ²	0.082	0.095	0.083	0.097
Adjusted R ²	0.082	0.095	0.083	0.096

Note:

*p<0.1; **p<0.05; ***p<0.01

PANEL B: NET REALIZED SPREAD

Dependent Variable Is Net Realized Spread (in bps) Using Individual Trades				
	(1)	(2)	(3)	(4)
ExchangeType	-0.136** (0.058)	0.003 (0.054)	-0.119** (0.058)	0.019 (0.054)
ParentOrderPOV	-15.913*** (1.985)	-10.208*** (1.797)	-17.364*** (1.906)	-11.697*** (1.725)
PctVolumeOverDailyTotal	-347.989*** (40.457)	-377.085*** (41.549)	-344.787*** (40.424)	-375.754*** (41.640)
BidAskSpreadInBps	0.012** (0.005)	0.058*** (0.007)	0.013** (0.005)	0.059*** (0.007)
LogTradePrice	1.101*** (0.107)	0.825** (0.377)	1.108*** (0.106)	0.738** (0.362)
LogTradeValue	-0.821*** (0.093)	-0.995*** (0.076)	-0.850*** (0.091)	-1.025*** (0.073)
TradeSide	-0.281* (0.159)	-0.326* (0.193)	-0.230 (0.159)	-0.253 (0.192)
TimeWindow FE	Yes	Yes	Yes	Yes
Stock FE		Yes		Yes
Day FE			Yes	Yes
SE CL	Stock and Day	Stock and Day	Stock and Day	Stock and Day
Observations	8,859,723	8,859,723	8,859,723	8,859,723
R ²	0.003	0.012	0.004	0.013
Adjusted R ²	0.003	0.011	0.004	0.013

Note:

*p<0.1; **p<0.05; ***p<0.01

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